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VIBRATION AND TEMPERATURE ANALYSIS OF A FOUR-POLE ELECTRIC MOTOR: SIGNAL FILTERING AND FREQUENCY-DOMAIN IDENTIFICATION OF MOTOR RUNNING SPEED, HARMONICS, AND THERMAL RESPONSE

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NOMENCLATURE

t	time, s
Δt	sampling period, s
F_s	sampling frequency, Hz
N	total number of sampled data points
k	discrete frequency index
f	frequency, Hz
f_k	discrete frequency at index k, Hz
Δf	frequency resolution, Hz
$a(t)$	time-domain vibration acceleration signal, g
$x(t)$	general measured time-domain signal
X_k	Fourier transform coefficient at index k
$A(f_k)$	single-sided vibration amplitude at frequency f_k, g
f_r	motor running-speed frequency, Hz
N_r	motor rotational speed, rpm
f_n	frequency of the nth harmonic, Hz
n	harmonic multiple
O	vibration order
A_{1x}	vibration amplitude at the running-speed frequency, g
$T(t)$	measured motor surface temperature, °C
$T[i]$	raw discrete temperature value at index i, °C
$T_{\text{filtered}}[i]$	filtered temperature value at index i, °C
$T_{\max}$	maximum filtered surface temperature, °C
M	moving-average window size, samples
m	moving-average summation index
τ_{avg}	average period of the aliased temperature oscillation, s
$t_{p,i}$	time location of the ith detected temperature peak, s
EMI	electromagnetic interference
DC	direct-current component or zero-frequency signal offset
FFT	Fast Fourier Transform

1. INTRODUCTION

This project presents vibration and temperature data collected from a four-pole electric motor in order to determine its dynamic and thermal behavior when it is being used. Using MATLAB, the raw vibrational signal was converted from a function of time to a function of frequency using the Fast Fourier Transform to determine the running speed, harmonic vibrational components, and any vibrational behavior below the running speed. The dominant vibrational frequency was found to be 59.4 Hz with an intensity of 2.467 g, ultimately corresponding to a motor speed of approximately 3564 rpm. The thermocouple temperature signal was filtered using a moving average to reduce aliased sinusoidal interference. From that filtered temperature response, we found that the motor surface reached a peak surface temperature of 45.44 °C. Collectively, the resulting plots and calculated values provide a clear representation of the motor's true vibrational and thermal behavior, while also highlighting potential sources of interference, such as electromagnetic interference.

2. MATERIALS AND METHODS

2.1 Materials

- MATLAB software for numerical computation and simulation
- A personal computer for code development and execution
- Project 3 motor test dataset containing vibration and temperature measurements
- Fast Fourier Transform method for frequency-domain vibration analysis
- Moving-average filtering method for temperature signal smoothing
- Sampling frequency of $F_s = 1000$ Hz
- Total sample length of $N = 5000$ data points
- Engineering references or scholarly sources for FFT analysis, vibration order analysis, and signal filtering methods

2.2 Discrete Fourier Transform

To evaluate the vibration characteristics of the motor, the Discrete Fourier Transform (DFT) was used to convert the vibration signal from the time domain into the frequency domain. This allowed the vibration dataset to be analyzed as a function of frequency instead of time. The frequency resolution and sampling parameters of the DFT depend on the sampling period, sampling frequency, and total number of samples:

Equation 2.2 $$X_k = \sum_{i=0}^{N-1} x[i] e^{-j2\pi ki/N} \quad [3]$$

where X_k represents the frequency-domain output at a frequency index k, $x[i]$ is the discrete time-domain signal at sample index i, N is the total number of data points, and j represents the imaginary unit [3].

2.3 Single-Sided Amplitude Spectrum

Because real-world signals produce a symmetrical spectrum around DC, the negative-frequency information produced by a two-sided DFT is unnecessary and redundant. To convert the two-sided spectrum into a single-sided amplitude spectrum, the negative half of the array is discarded, and the magnitude of every non-DC frequency is multiplied by two:

Equation 2.3 $$A(f_k) = 2 \times \left| \frac{X_k}{N} \right| \quad [1]$$

where X_k is the frequency-domain output at a frequency index k, N is total number of sampled-data points, f_k is the frequency value at index k, and A is the amplitude of the vibration signal at that frequency.

2.4 Vibration Frequency to RPM Conversion

To determine the running speed of the motor, we need to convert the frequency into revolutions per minute (RPM) using the following relationship:

Equation 2.4 $$N_r = 60 \times f_r$$ [4]

where f_r is the running vibration frequency of the motor (Hz) and RPM is the motor rotational speed in revolutions per minute.

2.5 Harmonic and Order Analysis

Vibration components that occur at exact multiples of the running speed are called harmonics or orders [4]. The frequencies of these harmonics and their corresponding orders are defined by:

Equation 2.51 $$f_n = n \times f_r$$ [4]

Equation 2.52 $$O = \frac{f}{f_r}$$ [4]

where f_n is the frequency of the nth harmonic, n is harmonic multiple, f is the specified frequency, f_r is the running-speed frequency, and O is the order.

2.6 Moving-Average Window Selection

To filter the noisy temperature dataset, we needed to use a moving-average, but in order to use a moving average, we needed an effective window to make sure we don't over or under smooth the data, so we wanted to find the period of the aliased sinusoidal. To find this, we looked for the difference in time between the peaks of the sinusoid and averaged them:

Equation 2.61 $$\Delta t_{p,i} = t_{p,i+1} - t_{p,i}$$

Equation 2.62 $$\tau_{avg} = \frac{1}{q-1}\sum_{i=1}^{q-1} \Delta t_{p,i}$$

where $\Delta t_{p,i}$ is the time difference between adjacent temperature peaks, $t_{p,i}$ is the time location of the i detected temperature peak, τ_{avg} is the average period of the aliased temperature oscillation, and q is the total number of detected peaks.

After finding the average time between each temperature peak, we convert it to a moving-average window size by multiplying it by the sampling frequency:

Equation 2.63 $$M = F_s \times \tau_{avg}$$

where F_s is the sampling frequency and M is the moving-average window size.

2.7 Temperature Moving-Average Filtering

Using the moving-average window size we found in 2.6, we apply it to the temperature signal being filtered by the moving average filter:

Equation 2.7 $$T_{filtered}[i] = \frac{1}{M}\sum_{m=0}^{M-1} T[i+m]$$ [2]

where $T_{filtered}[i]$ is the filtered temperature value at index i, $T[i+m]$ represents the current temperature sample when m=0 and the following temperature samples as m increases, i is the current sample index, m is the moving-average summation index, and M is the moving-average window size.

2.8 MATLAB Implementation

2.81 Loading Project 3 Data and Parameters

```
%% DATA LOADING AND SETTING UP PARAMETERS
load("C:\Users\jvocs\Downloads\Project_3_Dataset.mat")
```

This section loads the motor testing dataset into MATLAB and defines parameters and sets the sampling frequency Fs = 1000, the total number of sampled data points L=5000, and the sampling period T = 0.001. It also sets up the 5 second interval with 0.001 second increments. More importantly, the file also loads in the recorded motor temperature and vibration test samples.

2.82 Vibration Dataset Filtering

```
%% VIBRATION DATASET FILTERING
Y = fft(sigDist);
t = (0:L-1)*T;
P2 = 2*abs(Y/L);
P1 = P2(1:(L/2)+1);
f = Fs/L*(0:(L/2));
[maxstrength, idx] = max(P1);
dominantfrequency = (f(idx));
acceleration = bandpass(sigDist,[fix(dominantfrequency-2)
fix(dominantfrequency+2)],1000);
RPM=60*dominantfrequency;
```

This section analyzes the vibration dataset in the frequency domain. First, the vibration signal sigDist is transformed using the Fast Fourier Transform, as we've seen in 2.2, as it converts the vibration data from a time-based signal into a frequency-based signal. The FFT output is then scaled and converted into a single-sided amplitude spectrum P1, which shows the vibration strength at each frequency. The vector f is created so that each amplitude in P1 corresponds to a frequency in Hz, ultimately setting up the frequency domain. Then, the strongest amplitude in the spectrum is found using max(P1), and the index of that maximum value is used to find the dominant vibration frequency. This dominant frequency is treated as the motor running-speed frequency. A bandpass filter is then applied to the raw vibration signal using a range of approximately two hertz below and above the dominant frequency, isolating the vibration behavior near the running speed and stores the filtered signal as acceleration. Finally, the dominant frequency is converted to revolutions per minute using the relationship displayed in 2.4.

2.83 Temperature Signal Filtering

```
%% TEMPERATURE SIGNAL FILTERING
[peaks,tmaxs] = findpeaks(TC_Noise,t);
peaktopeaktimes = diff(tmaxs);
avgperiod = mean(peaktopeaktimes);
k = fix(Fs*avgperiod);
temps = movmean(TC_Noise,k);
peaktemperature = max(temps);
```

This section filters the noisy temperature signal. To do this, we apply the methods in 2.6. The function findpeaks is first used to identify the local maxima across the raw temperature signal TC_Noise, along with the time locations of those peaks. The differences between consecutive peak times are then calculated using diff(tmaxs), giving the approximate period of each oscillation in the noisy temperature signal. These peak-to-peak times are then averaged using mean(peaktopeaktimes) so that we can estimate the average period of the aliased sinusoidal interference. The average period is then converted using equation 2.63. The fix function is used so that the window size becomes a whole number and can be used as a window size. The moving average filter (eqn 2.7) is then applied to the raw temperature data using movmean(TC_Noise,k), which produces the filtered temperature signal as temps. Finally, the maximum value of the filtered temperature signal is found and stored as peaktemperature.

2.84 Creating Figures and Plotting Data

```
%% CREATING FIGURES AND PLOTTING DATA
figure(1)
subplot(2,1,1)
plot(t,sigDist)
title("Raw Amplitude (g) vs. Time (s)")
xlabel("Time(s)")
ylabel("Amplitude (g)")
grid on
subplot(2,1,2)
plot(t,TC_Noise)
title("Raw Temperature (C) vs. Time (s)")
xlabel("Time(s)")
ylabel("Raw Temperature (C)")
grid on
```

This section generates the raw signals. figure(1) contains two subplots showing the original unfiltered datasets. The first subplot displays the raw vibration amplitude over time while the second subplot displays the raw temperature signal over time.

```
figure(2)
subplot(2,1,1)
plot(t,acceleration)
title("Filtered Amplitude (g) vs. Time (s)")
xlabel("Time(s)")
ylabel("Filtered Amplitude (g)")
grid on
subplot(2,1,2)
plot(t,temps)
title("Filtered Temperature (C) vs. Time (s)")
xlabel("Time(s)")
ylabel("Filtered Temperature (C)")
grid on
```

This section generates the filtered signals. figure(2) contains two subplots showing the resulting filtered datasets. The first subplot displays the bandpass-filtered vibration amplitude over time while the second subplot displays the moving-average filtered temperature signal over time.

```
figure(3)
plot(f,P1)
title("Amplitude (g) vs. Frequency (Hz)")
xlabel("Frequency (Hz)")
ylabel("Amplitude (g)")
grid on
```

This section generates the amplitude vs frequency spectrum. This plot is used to identify the dominant vibration frequency, harmonic components, and other vibration behaviors.

Each plot is labeled with a title, has appropriate y and x axis labeling, and has a grid on to properly and clearly display the data.

2.85 Displaying Data

```
%% DISPLAYING DATA
disp("The motor is spinning at "+RPM+" revolutions per
minute, as it has a vibration frequency of
"+dominantfrequency+" Hz with an intensity of
"+maxstrength+" g.")
disp("The peak temperature reached is "+peaktemperature+"
degrees celsius.")
```

This section displays the main calculated results in the command window. The first display statement displays the calculated motor speed in revolutions per minute, the dominant vibration frequency in hertz, and the vibration intensity at that frequency in g's. The second display statement displays the maximum filtered temperature value, which represents the peak motor surface temperature reached during the test.

2.86 Tabulating Vibration Order and Harmonic Data

```
%% TABULATING VIBRATION ORDER AND HARMONIC DATA
foi = [29.6,59.4,118.6,178,237.4,296.6,356,415.4,474.6];
ooi = round(foi/dominantfrequency,1);
idx = round(5*foi + 1);
voi = P1(idx);
ovoi = voi/maxstrength;
tbl = table(foi', ooi', voi', ovoi', 'VariableNames',
{'Frequency (Hz)','Order','Vibration (g)','Running-Speed
Comparison'});
disp(tbl)
```

This section organizes the vibration order and harmonic data into a table. The array foi stores the frequencies of interest that were identified from the vibration spectrum. These frequencies include the below running speed component, the running speed frequency, and the main harmonic components. The order of each frequency of interest is calculated by dividing each frequency of interest by the dominant running-speed frequency (eqn 2.52). The result is then rounded to the nearest tenth to get whole numbers and preserve any sub-running-speed components.

The index values corresponding to each frequency of interest are then calculated using idx = round(5*foi+1), which works because the frequency resolution is 0.2 Hz, so each frequency step corresponds to five index positions per hertz, with one added to account for array indexing. The vibration amplitude at each frequency of interest is extracted from the amplitude spectrum and stored in voi. Each vibration amplitude is then divided by the running speed vibration amplitude to create ovoi, which compares each selected vibration to the strongest vibration (running speed).

3. DATA

3.1 Sampled Data, Parameters, and Results

Sampling Frequency: 1000 Hz

Sampling Period: 0.001 s

Total Number of Samples, L,N: 5000 samples

Test Duration: 5 s

Dominant Vibration Frequency, fr: 59.4 Hz

Motor Running Speed, Nr: 3564 rpm

Running-Speed Vibration Intensity: 2.467 g

Peak Filtered Temperature: 45.44 °C

Frequencies of Interest: 29.6 Hz, 59.4 Hz, 118.6 Hz, 178 Hz, 237.4 Hz, 296.6 Hz, 356 Hz, 415.4 Hz, 474.6 Hz

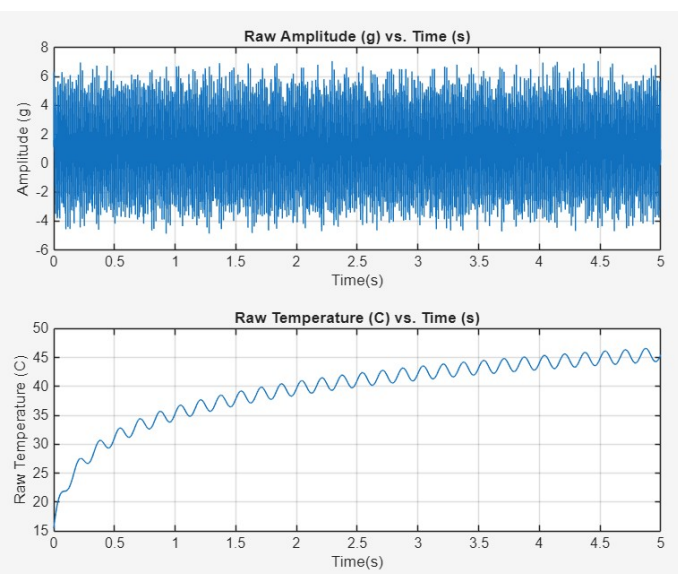

FIGURE 3.10: RAW VIBRATION AND TEMPERATURE SIGNALS OVER TIME

3.2 Figures, Models, and Tables

Table 3.21			
Frequency (Hz)	Order	Vibration (g)	Relative Amplitude
29.6	0.5	0.114	0.046
59.4	1.0	2.467	1.000
118.6	2.0	0.558	0.226
178.0	3.0	0.322	0.131
237.4	4.0	1.244	0.504
296.6	5.0	0.746	0.302
356.0	6.0	0.604	0.245
415.4	7.0	0.241	0.098
474.6	8.0	0.415	0.168

TABLE 3.21: FREQUENCY, ORDER, AND VIBRATION AMPLITUDE SUMMARY

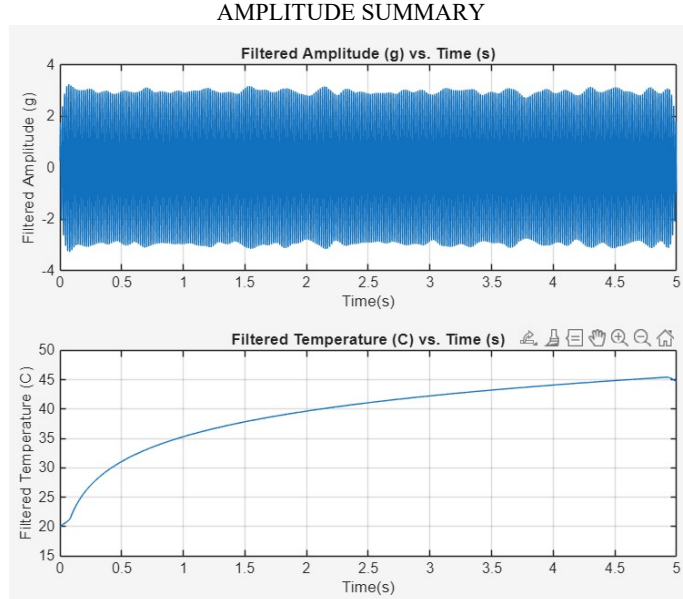

FIGURE 3.22: FILTERED MOTOR VIBRATION AND TEMPERATURE RESPONSES OVER TIME

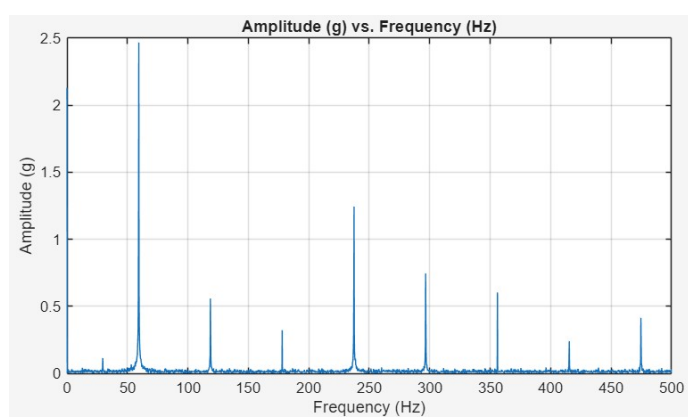

FIGURE 3.23: FFT VIBRATION SPECTRUM OF MOTOR TEST DATA

4. RESULTS

Using the FFT conversion, we found the motor's dominant vibrational frequency to be 59.4 Hz with an intensity of 2.467 g, which corresponds to a motor running speed of 3564 rpm using the conversion factor from eqn. 2.4. The filtered temperature signal, on the other hand, reached a peak surface temperature of 45.44 °C. Moreover, the vibration spectrum generated by the FFT conversion displayed several harmonic components at multiples of the running-speed frequency, including a 2x harmonic at 118.6 Hz with an intensity of 0.556 g and a 4x harmonic at 237.4 Hz with an intensity of 1.244 g. A sub-running-speed component was also identified at 29.6 Hz, about half of the running speed, with an intensity of 0.114 g. Overall, the strongest vibration response occurred at the running speed, while the 4x harmonic was the strongest higher-order harmonic.

5. DISCUSSION

5.1 Interpretation of Running Speed and Harmonics

The FFT analysis showed that the dominant motor vibration frequency was 59.4 Hz, corresponding to the running-speed frequency because it had the strongest vibration peak in Figure 3.23, at 2.467 g. A harmonic is a vibration component that occurs at a multiple of the running-speed frequency [4]. For example, if the running speed is f_r, then the 2x harmonic should be expected at $2f_r$, the 3x harmonic at $3f_r$, and the 4x harmonic at $4f_r$. The measured 2x and 4x harmonics were at 118.6 Hz (0.558 g) and 237.4 Hz (1.244 g), respectively, which closely matched the expected values. However, the 4x harmonic was significantly stronger compared to the 2x harmonic, reaching approximately 50.4% of the 1x amplitude, ultimately indicating that while the dominant vibrational source is shaft rotation at the running speed, another vibration component also exists at four times the running speed.

5.2 Sub-Running-Speed and Higher-Order Vibration

A sub-running-speed vibrational frequency was identified at 29.6 Hz, corresponding to a 0.5x order. This component had an intensity of 0.114 g, which is only about 4.6% of the running-speed intensity. Since this response was smaller than the dominant vibration, it was present in the amplitude spectrum but not a dominant frequency. The higher-order components from 3x to 8x, while having lower intensities than the running speed, demonstrate that the vibrational signal contained a repeating pattern associated with motor rotation.

5.3 Temperature Response and Filter Effectiveness

The raw temperature signal contained a sinusoidal disturbance, making the true temperature difficult to identify. A moving-average filter was applied to smooth the signal and reduce the aliased sinusoidal noise. After filtering, we obtained a clearer thermal response plot for analysis, revealing a peak motor surface temperature of 45.44 °C. Without filtering, however, the peak temperature could have been affected by noise rather than representing the true underlying motor surface temperature.

5.4 Noise Sources

Electromagnetic interference (EMI) can be a source of noise in measured data, as electrical or magnetic fields from nearby equipment can affect the sensor signal or wiring. Since the test involved an electric motor, the motor's electrical current, magnetic field, or nearby lab equipment could have introduced unwanted electrical noise into the vibration and temperature datasets.

Clear harmonic peaks at multiples of the running speed (59.4 Hz) were interpreted as genuine mechanical vibration rather than interference. While the smaller peak at 29.6 Hz (0.5x order) may initially resemble EMI, EMI typically produces broadband noise or peaks unrelated to shaft speed, so that is unlikely. In contrast, the 0.5x component occurs at a precise sub-harmonic, exactly half the running speed. This is a well-known characteristic of mechanical looseness, where a loose component is excited only every other revolution of the shaft [5].

According to industry guidelines, mechanical looseness often produces a 0.5x sub-harmonic along with a series of higher-order harmonics [5]. This matches the observed spectrum well. Additionally, the dominant 1x peak at 59.4 Hz is consistent with rotor unbalance, while the strong peaks at 2x (118.6 Hz) and 4x (237.4 Hz) orders suggest possible shaft misalignment [5]. Collectively, these anomalies indicate that the motor is experiencing a combination of unbalance, misalignment, and mechanical looseness rather than pure signal noise.

6. CONCLUSION

This study analyzed vibration and temperature data from a four-pole electric motor using FFT-based frequency analysis and signal filtering methods. The FFT identified a dominant running-speed frequency of 59.4 Hz (3564 rpm) with an amplitude of 2.467 g. Further harmonic analysis revealed a 2x harmonic at 118.6 Hz (0.558 g), a 4x harmonic at 237.4 Hz (1.244 g), and a 0.5x sub-harmonic at 29.6 Hz (0.114 g). Rather

than being random noise, according to industry guidelines, these specific peaks point to distinct physical faults, with the 1x dominant frequency indicating rotor unbalance, the strong 2x and 4x peaks indicates misalignment, and the 0.5x sub-harmonic indicating mechanical looseness [5]. For the temperature dataset, however, had to be filtered using a moving-average method to reduce the sinusoidal interference within the signal. After filtering, the peak motor surface temperature was found to be approximately 45.44 °C. Overall, the FFT and filtering methods separated the main motor vibration and temperature behavior from the noise and electromagnetic interference. The analysis demonstrates the importance of filtering, because raw data can contain misleading noise that hides the true behavior of the signal.

7. REFERENCES

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